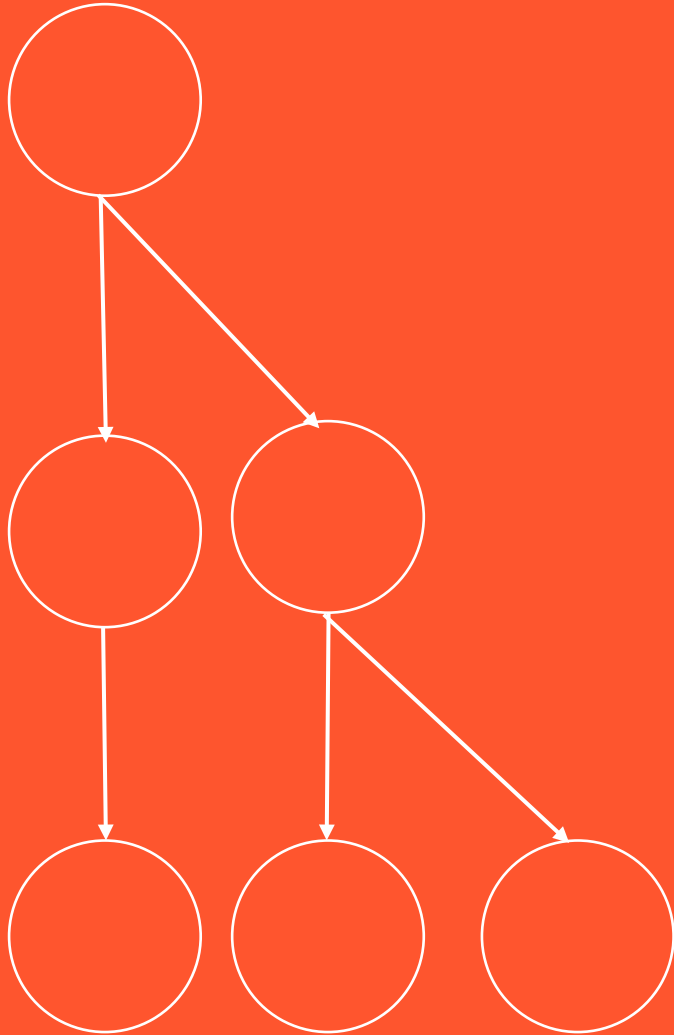


# BCOG 100

Spatial Cognition

Lecture 3: Neurobiological Implementation



UNIVERSITY OF  
**ILLINOIS**  
URBANA-CHAMPAIGN

## Reading Check 8-3

**Password:** hippocampus

**Question:** Do you have good or bad spatial cognition? Tell me about it.

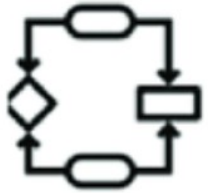


# Neurobiology of Spatial Cognition



## Computation

*What does the system do?*



## Algorithm

*How does the system do it?*



## Implementation

*How is the system realized?*

## Recap & Overview

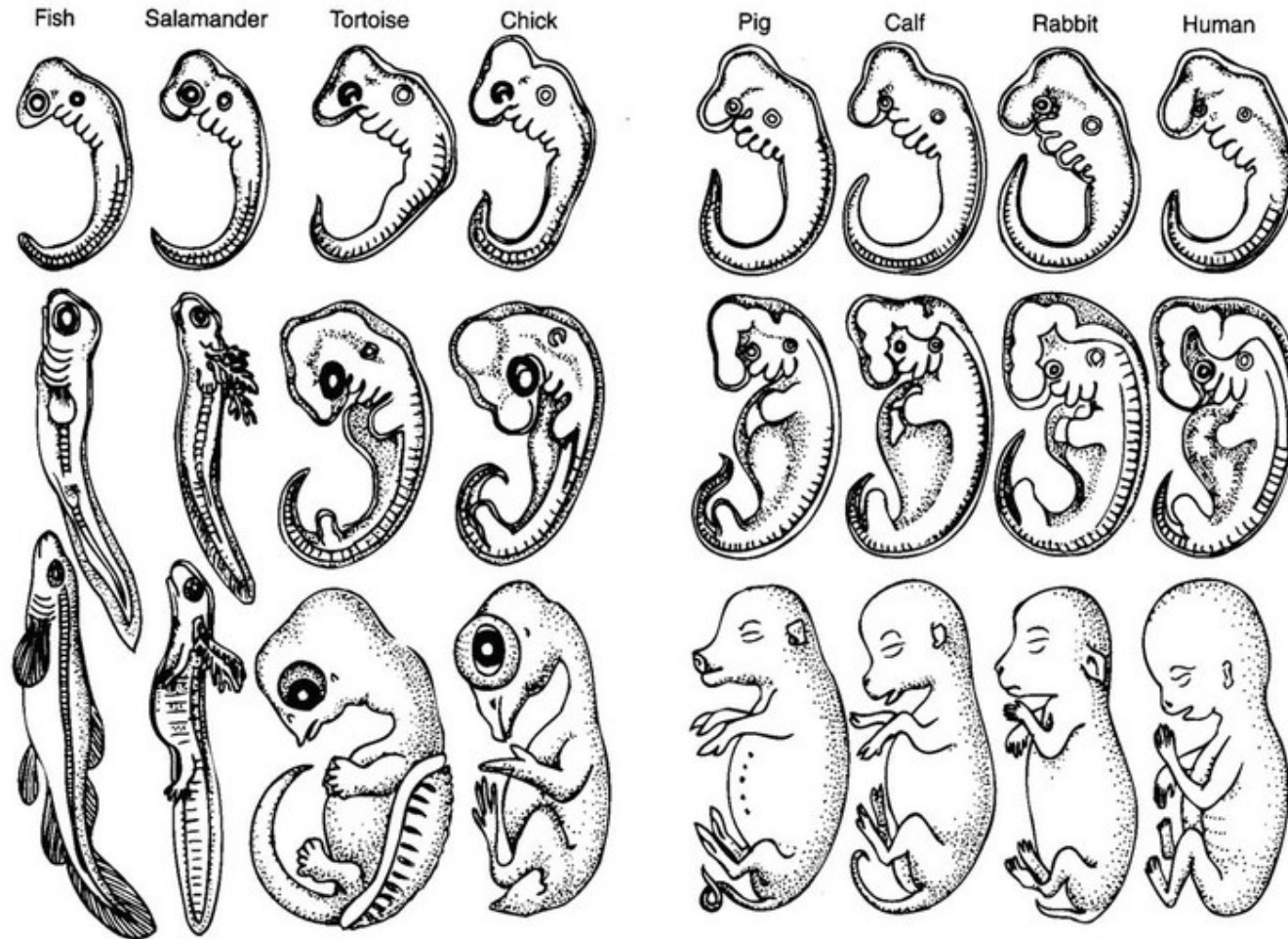
### Last Time:

Representation and Algorithms for Spatial Cognition

### Today:

- From abstract models to neural systems
- How brains build and update internal maps
- How spatial circuits evolved into memory systems

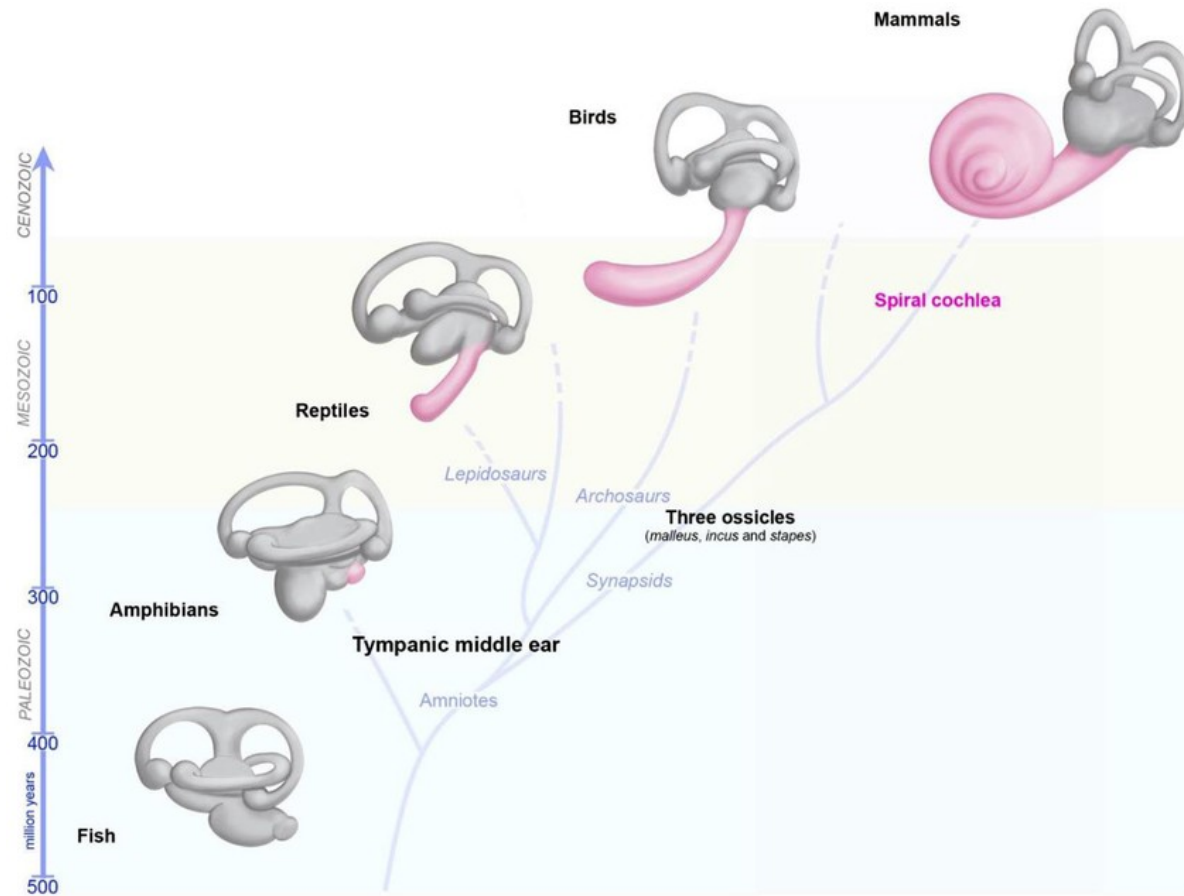
# Neurobiology of Spatial Cognition



## Evolutionary Continuity: From Fish to Mammals

- Spatial cognition builds on ancient orientation systems
- Early vertebrates integrated motion and sensory cues
- Modern hippocampal circuits extend those same principles

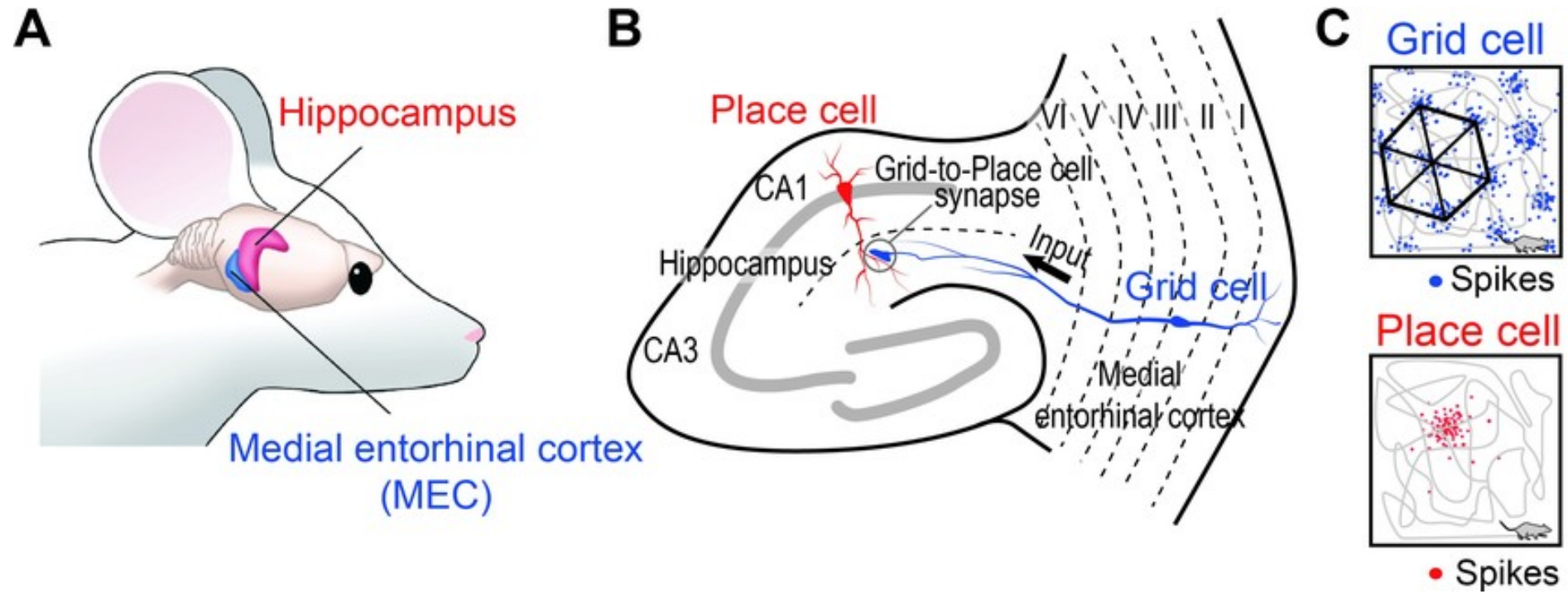
# Neurobiology of Spatial Cognition



## Early Neural Foundations: Stabilizing Movement and Orientation

- Vestibular, cerebellar, and brainstem circuits integrate motion signals
- Different senses must agree for stable orientation
- Foundation for distinguishing self-motion from world motion

# Neurobiology of Spatial Cognition

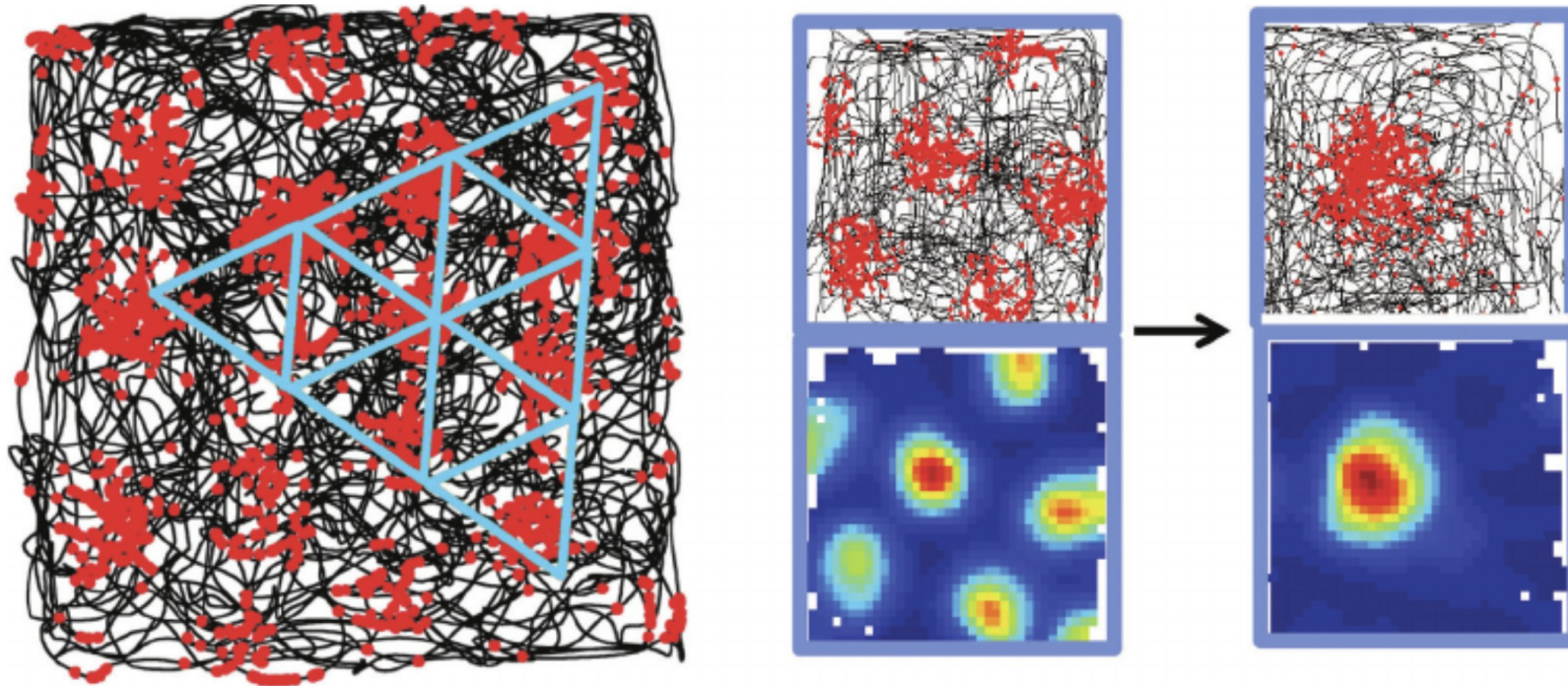


## Place Cells: The Neural Map of Space

- Vestibular, cerebellar, and brainstem circuits integrate motion signals
- Different senses must agree for stable orientation
- Foundation for distinguishing self-motion from world motion



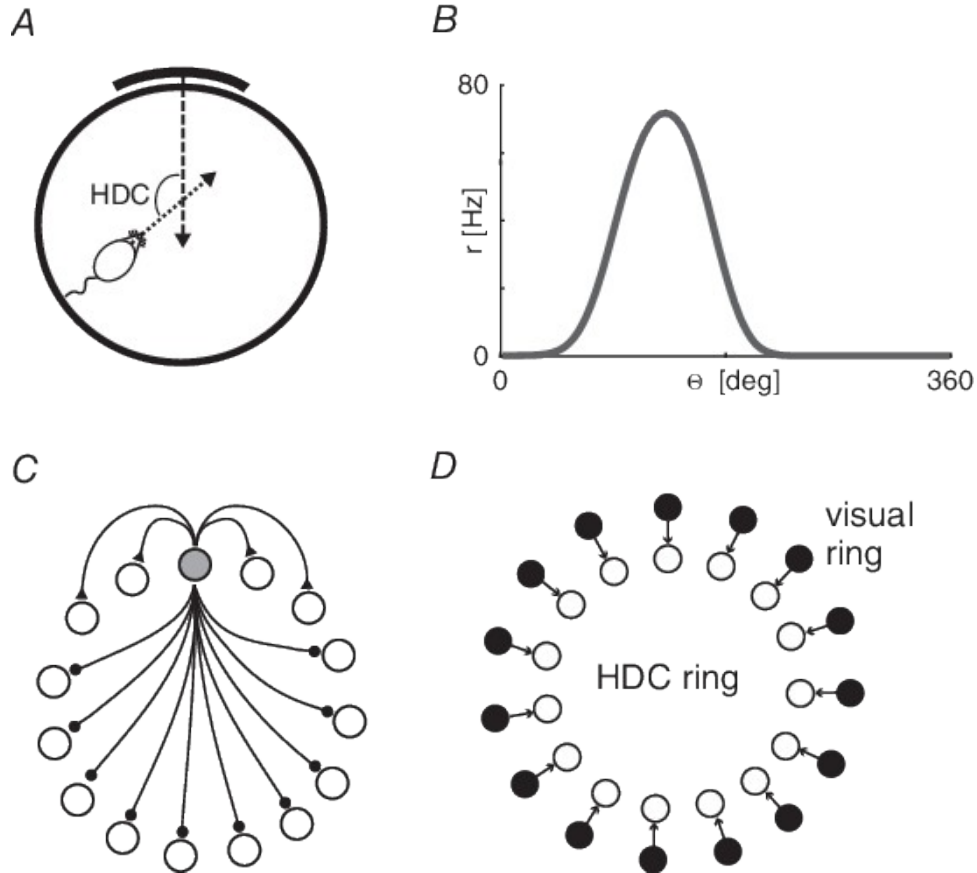
# Neurobiology of Spatial Cognition



## Grid Cells: The Brain's Internal Coordinate System

- Moser & Moser (2005): entorhinal neurons fire in hexagonal patterns
- Each grid cell marks locations at regular spatial intervals
- Provides a built-in metric for distance and direction

# Neurobiology of Spatial Cognition

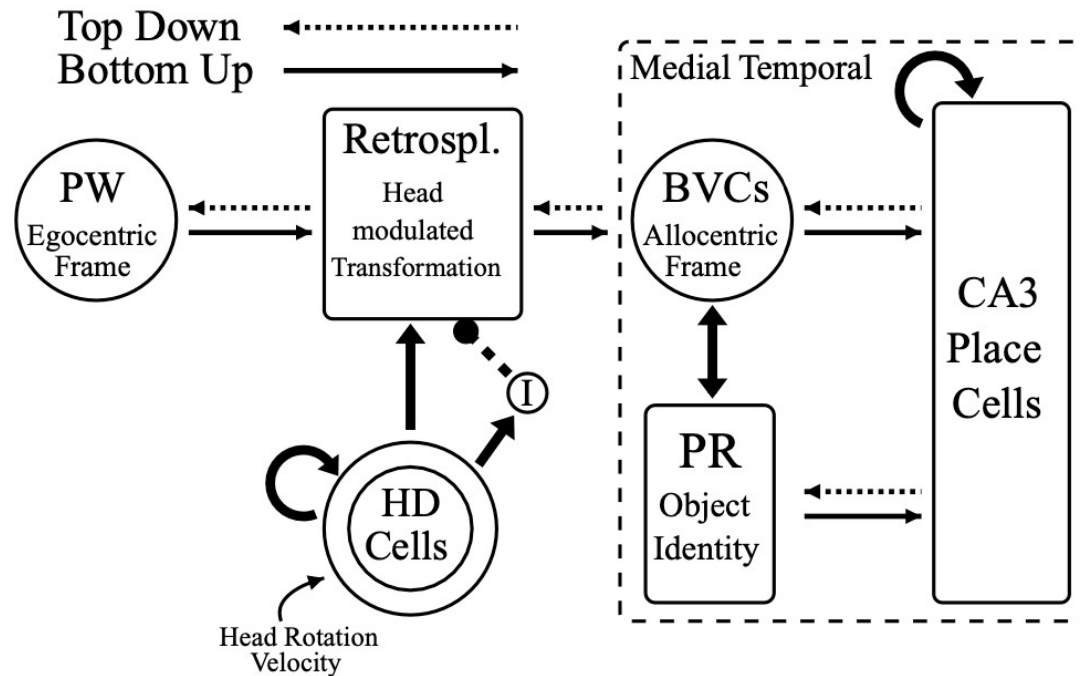


## Head-Direction and Boundary Cells: Orientation and Limits

- Head-direction cells encode facing orientation
- Boundary cells fire near environmental edges
- Together they anchor spatial maps to stable reference frames



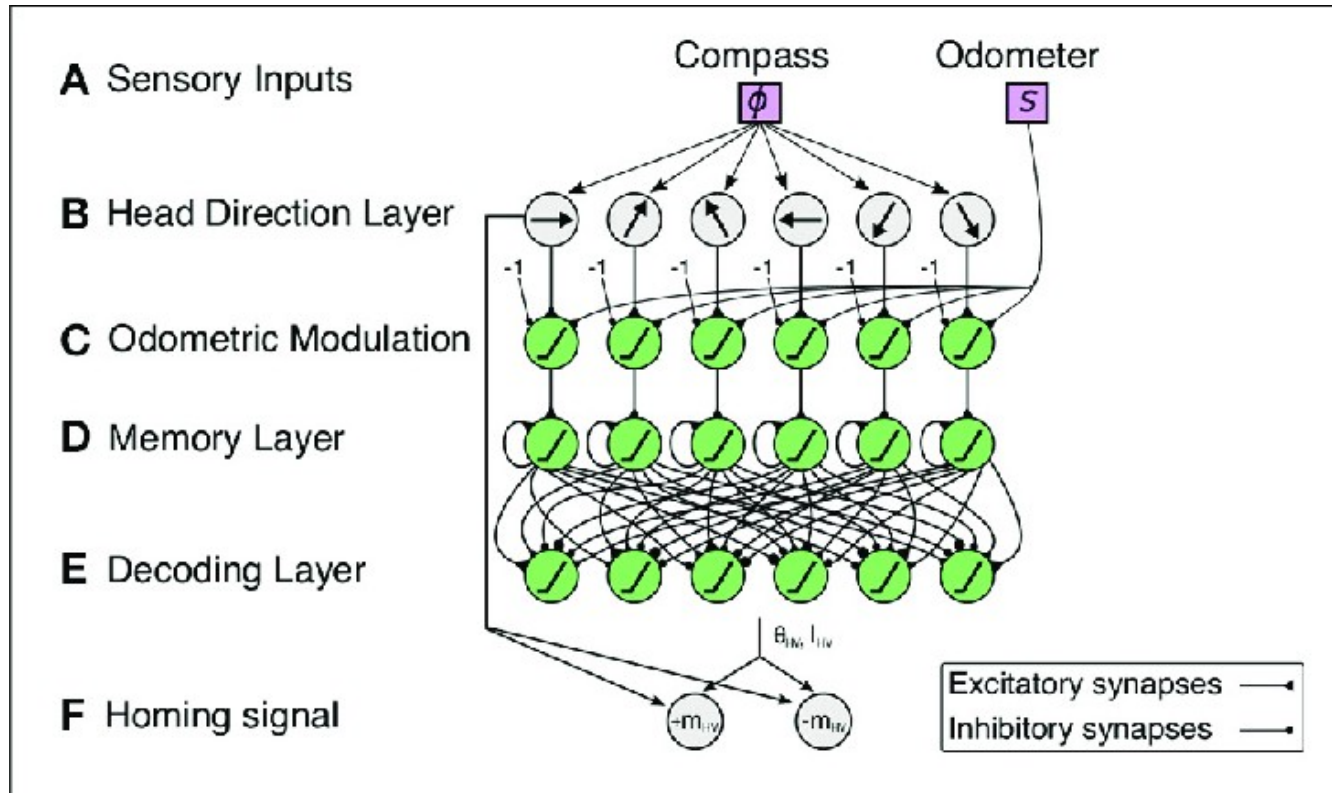
# Neurobiology of Spatial Cognition



## Interacting Systems: Building a Coherent Spatial Map

- Hippocampus and entorhinal cortex form a shared navigation circuit
- Place, grid, head-direction, and boundary cells work together
- Integration creates a stable, self-correcting map of space

# Neurobiology of Spatial Cognition



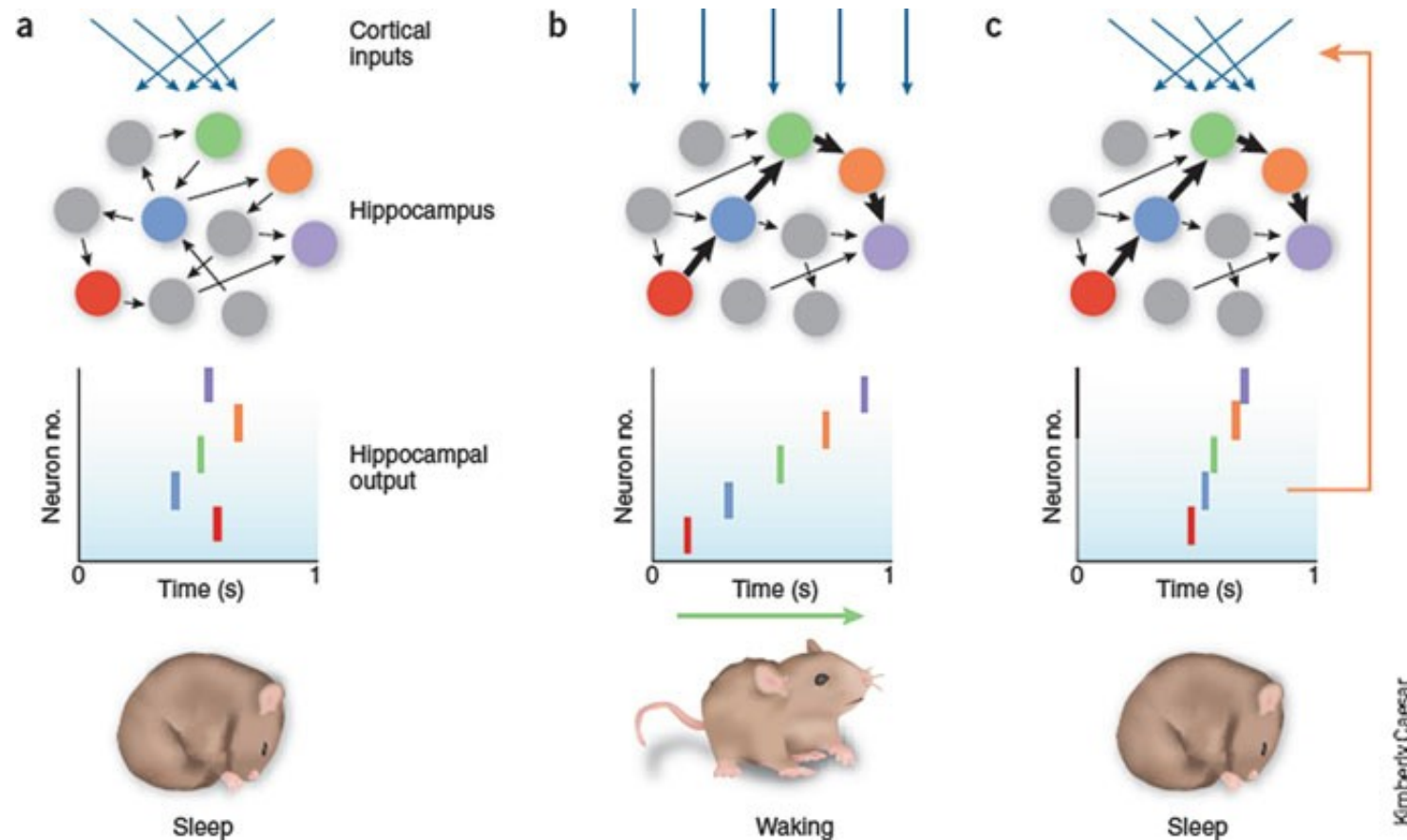
## Neural Mechanisms of Path Integration

- Grid and head-direction cells track movement through space
- Velocity and orientation signals update the map
- Landmarks recalibrate drifting internal estimates

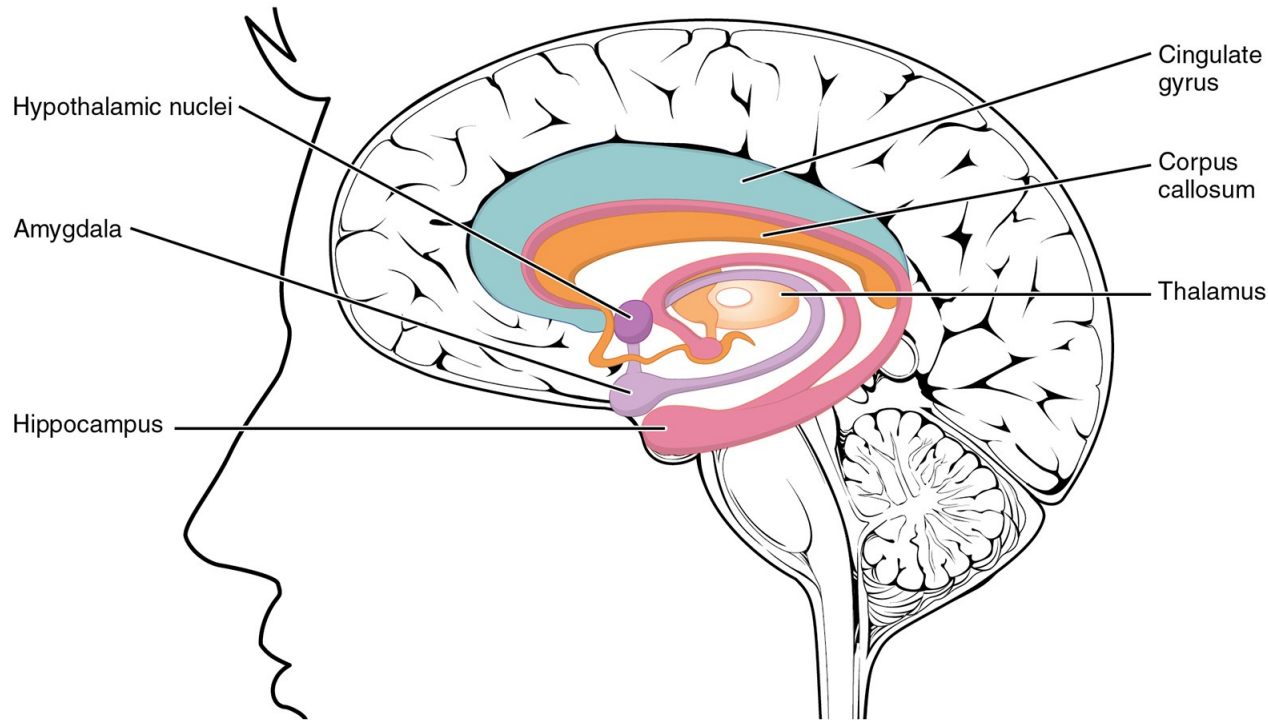
# Neurobiology of Spatial Cognition

## Replay and Pre-play: Planning and Prediction in the Hippocampus

- Hippocampal “place cells” reactivate during rest and sleep
- Replay: past trajectories repeated in fast sequence
- Pre-play: upcoming paths simulated before movement



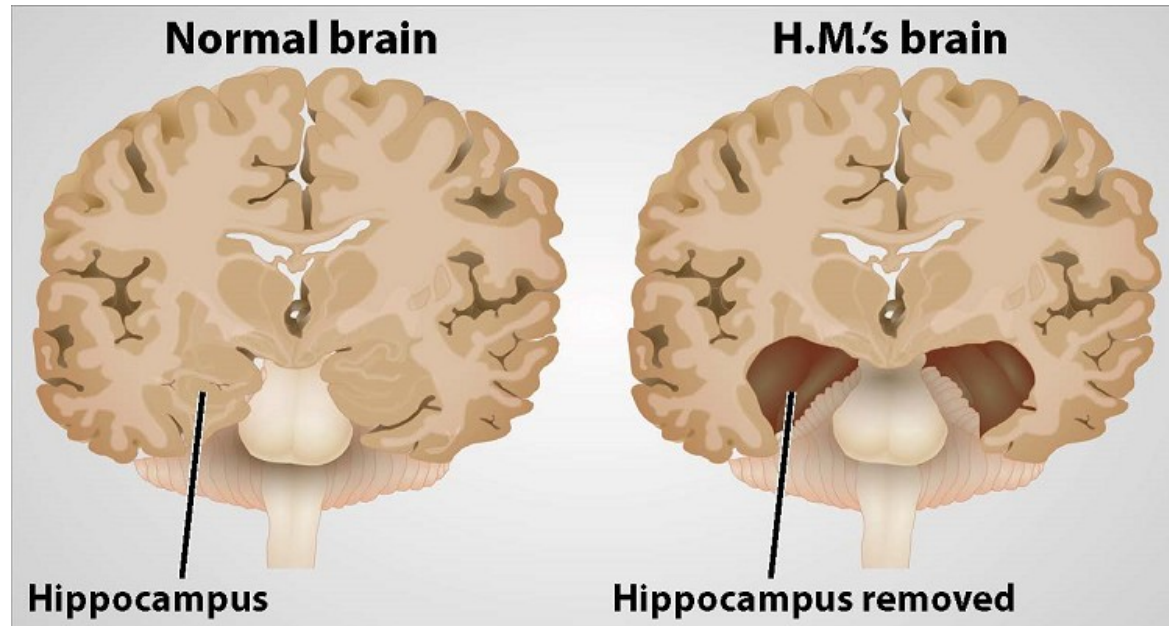
# Neurobiology of Spatial Cognition



## From Navigation to Memory

- Spatial mapping and episodic memory share the same circuitry
- The hippocampus links *where*, *when*, and *what happened*
- Navigation through space became navigation through experience

# Neurobiology of Spatial Cognition

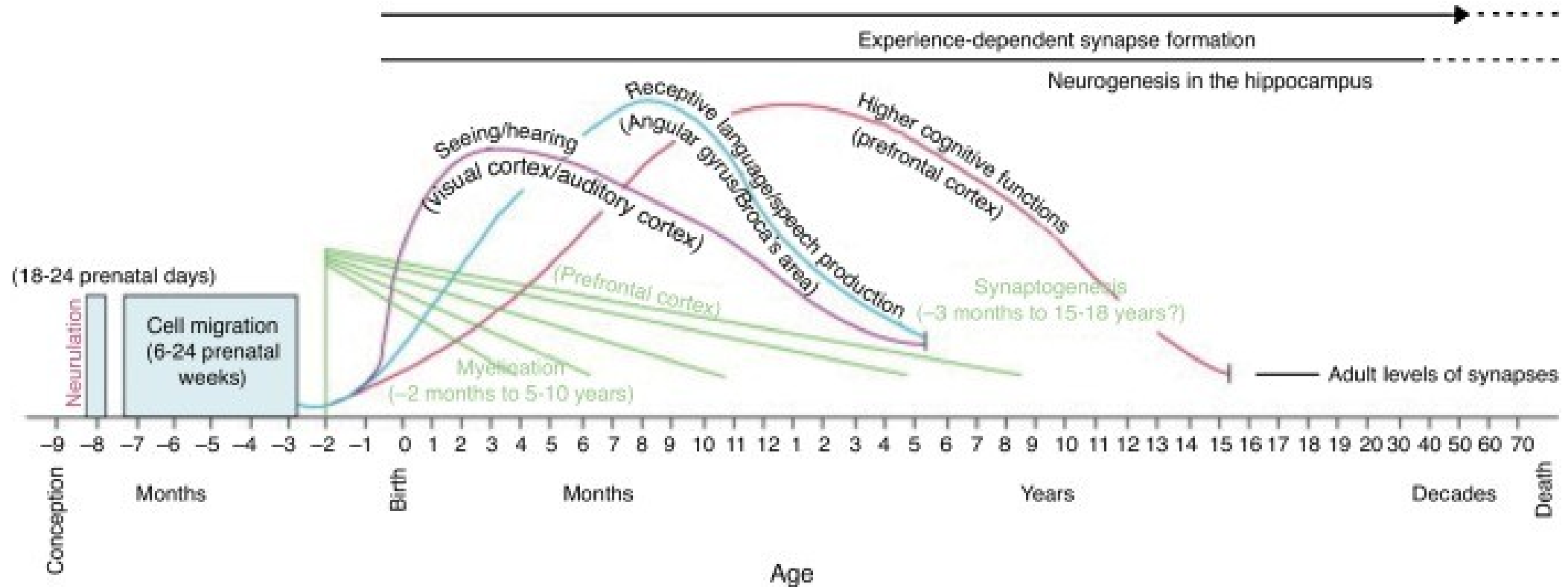


## Lesions and Imaging Evidence

- Hippocampal damage impairs both navigation and episodic memory
- Imaging shows spatial and memory tasks activate overlapping regions
- Reactivation reinstates the context of past experience



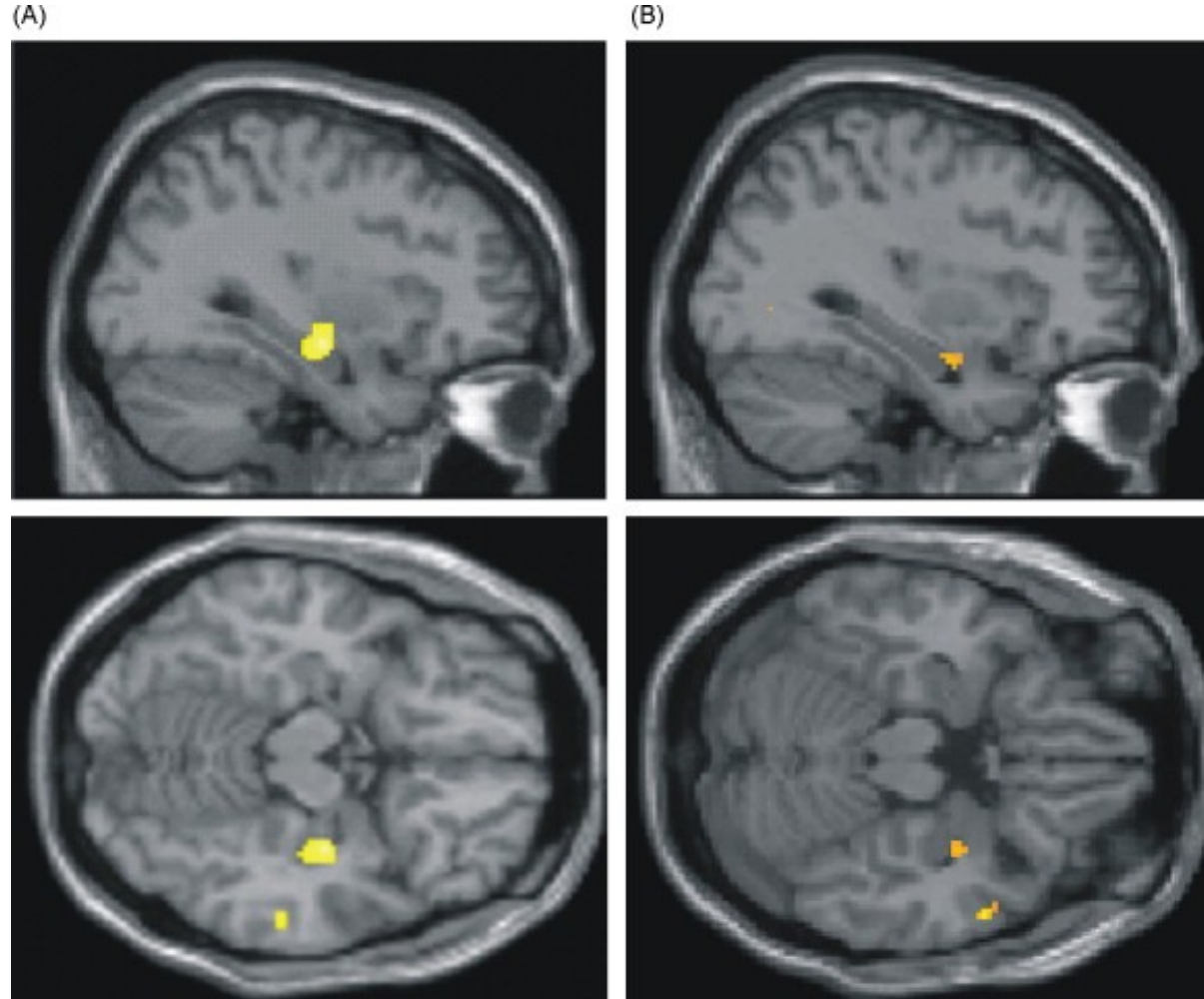
# Neurobiology of Spatial Cognition



## Development of Spatial Maps

- Children's spatial abilities and episodic memory develop in parallel
- Allocentric mapping emerges gradually across early childhood
- Reflects hippocampal maturation and experience-driven learning

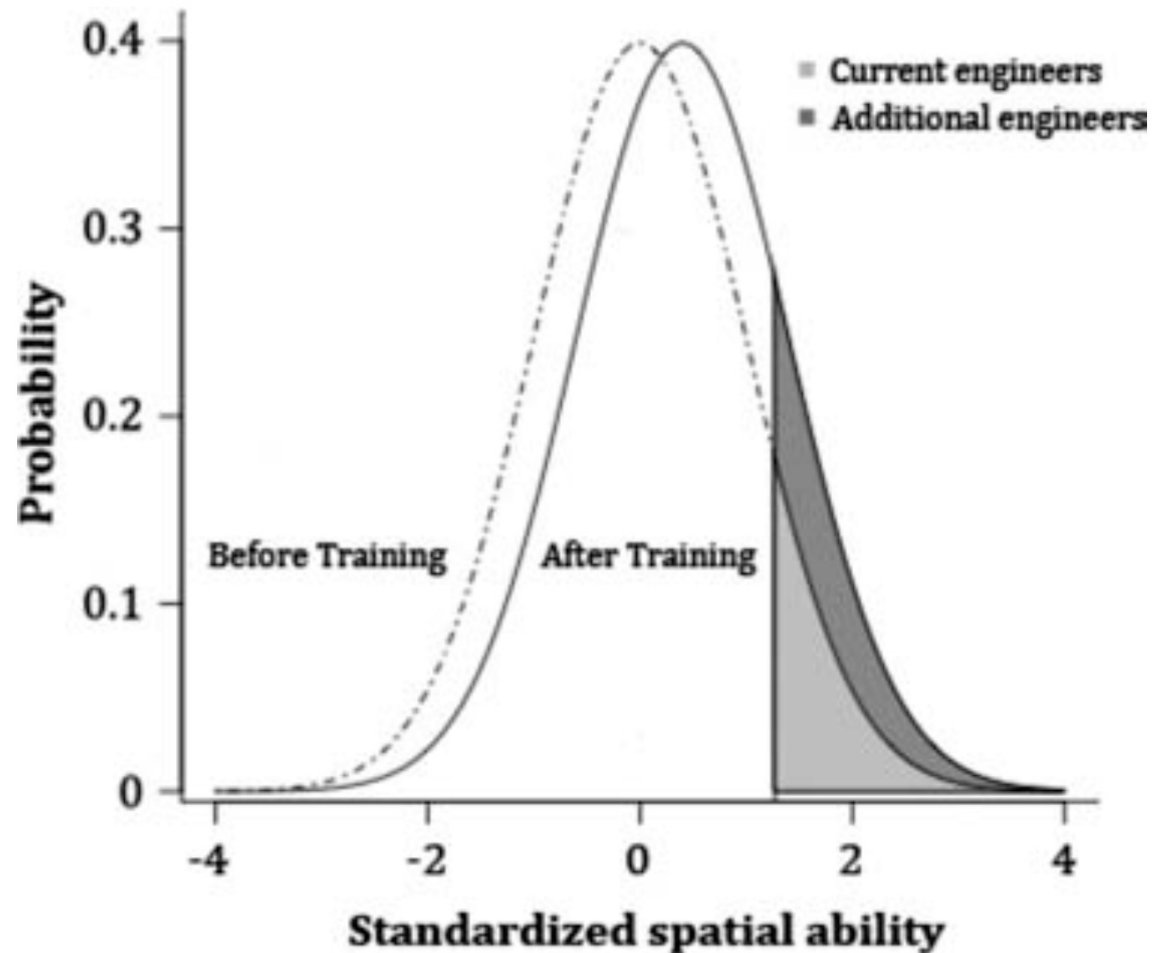
# Neurobiology of Spatial Cognition



## Plasticity and Experience

- Children's spatial abilities and episodic memory develop in parallel
- Allocentric mapping emerges gradually across early childhood
- Reflects hippocampal maturation and experience-driven learning

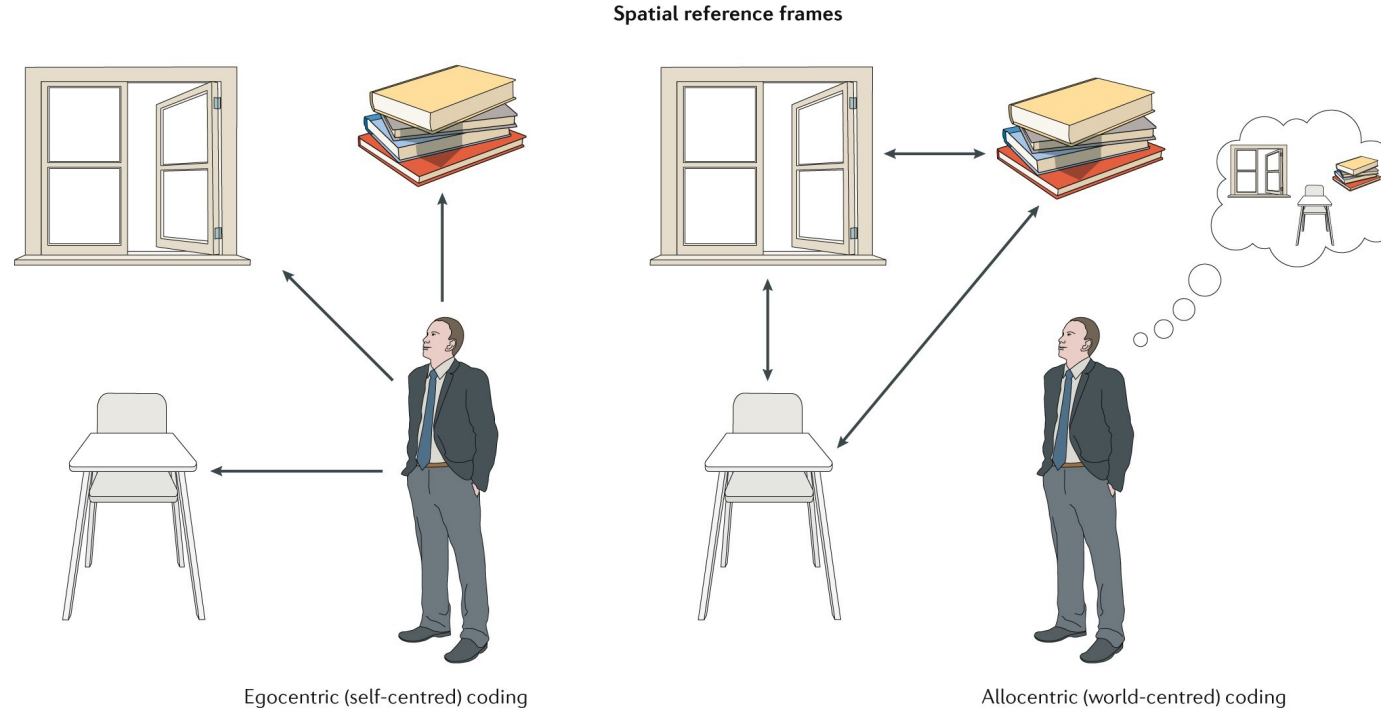
# Neurobiology of Spatial Cognition



## Can Spatial Ability be Improved?

- Uttal et al (2013) show spatial skills *can* be improved through training
- Benefits transfer across tasks and persist over time
- Most effective programs involve active, varied practice

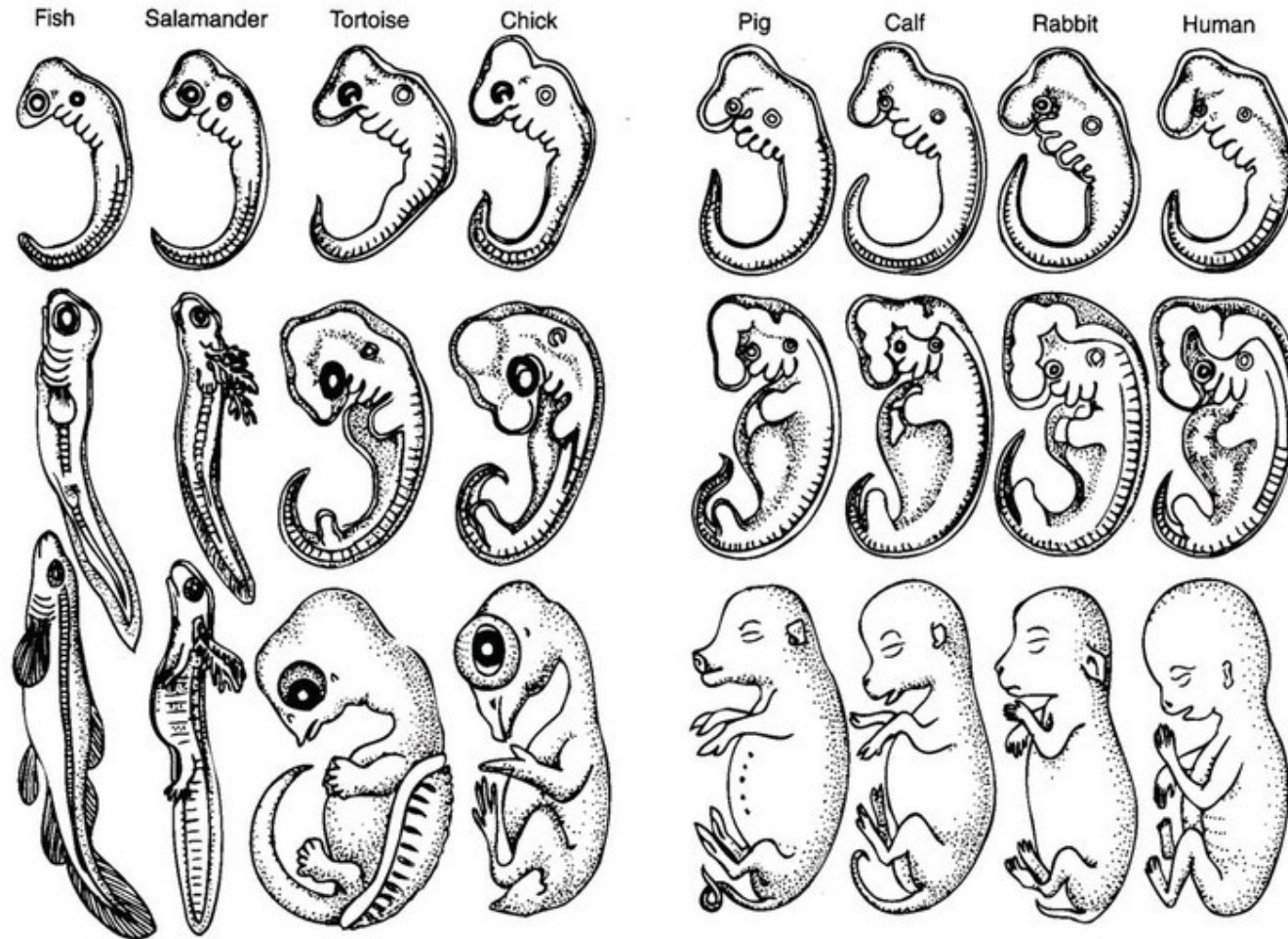
# Neurobiology of Spatial Cognition



## Disorders of Spatial Cognition

- Damage to hippocampal and parietal systems disrupts navigation
- Topographical disorientation: intact perception, lost “sense of place”
- Spatial deficits reveal how maps anchor memory and self-location

# Neurobiology of Spatial Cognition



## Comparative and Clinical Insights

- Spatial mapping systems are evolutionarily conserved
- Different species adapt shared circuits for unique navigation needs
- Comparing across species clarifies what's fundamental vs. derived



# Neurobiology of Spatial Cognition

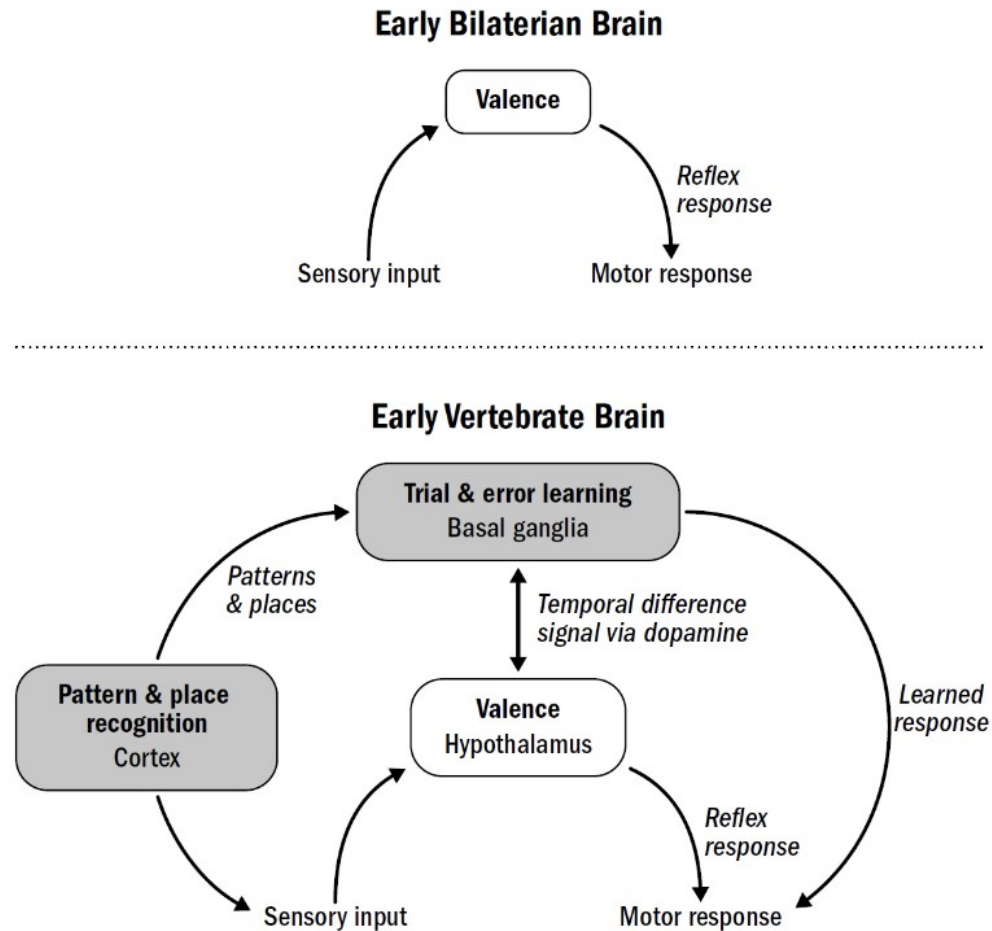


Figure 9.3

Original art by Rebecca Gelernter

## Spatial Maps and Predictive Brains

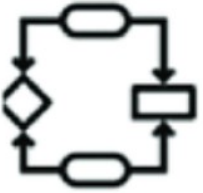
- Spatial mapping may be the evolutionary origin of prediction
- Maps let brains simulate outcomes before acting
- From navigating space to navigating possibility

# Spatial Cognition and Curiosity



## Computation

*What does the system do?*



## Algorithm

*How does the system do it?*



## Implementation

*How is the system realized?*

## Recap

- Spatial cognition evolved to support predictive navigation
- Cognitive and neural maps encode relations, not just places
- Mapping space laid the groundwork for mapping experience

## Up Next

- How the brain maps time and experience — Memory
- Spatial cognition as the bridge between perception and thought